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Climbing abilities in four species of desert rodents

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The potential locomotor performances of four Saharan species of rodents were tested in laboratory conditions, using three experimental situations. Two questions were investigated: Is each of these species able to use the vertical dimension when it is available? Does each of these species behave efficiently in a new environment? We tested the performances of *Acomys cahirinus* (Desmarest 1819) and *Gerbillus campestris* (Le Vaillant 1867) that are found in rocky environments in the wild and of *Gerbillus pyramidum* (I. Geoffroy 1825) and *Meriones shawi* (Rozet 1833), that are found in sandy environments. Climbing performance of the species was tested on a horizontal platform, a thread and a tree-branch. From our data, we concluded that *Acomys* and *G. campestris* showed an efficient behaviour in these three tasks. *M. shawi* and to a lesser extent *G. pyramidum* were unable to behave efficiently in the test situations.

KEY WORDS: locomotor performances, desert rodents, Sahara.

INTRODUCTION

All species do not exhibit the same behavioural abilities to move easily and efficiently in an environment. Vertebrate species are not evenly distributed in the Saharan desert (LE BERRE & LE GUELTE 1989). Many species are restricted to mountains and only 40% of species are common to plains and mountains.

Some species of rodents are more proficient than others in locomotion in more or less complex environments. Some species are strongly bound to ground locomotion, such as guinea pigs, marmots (*Marmota marmota* Linnaeus 1758) or gerbils. Others are very skilled in rocky environments (gundis), others are tree-dwelling animals (squirrels, dormice) or can swim (beavers, coypus). Swimming abilities of different species have been extensively compared (HICKMAN et al. 1983, HICKMAN & MACHINÉ 1986).

Studies of geographic variation in the tree-climbing ability and external morphology of rodents of the genus *Peromyscus* Gloger 1841 (HORNER 1954, LAYNE 1970) have shown that spatial variation in natural selection influences both behaviour and morphology. Individuals of a given species living in wooded areas have greater climbing abilities and longer feet than those of individuals of the same species living in shrubland. THOMPSON (1990), who has studied two subspecies of *Peromyscus*

maniculatus (*P. m. rufinus* Merram 1890, and *P. m. sonoriensis* Le Conte 1853) showed that individuals of the first subspecies living in conifer forests and woodland in Arizona (USA) are better climbers than individuals of the second subspecies living either in conifer woodland or desert habitats in Nevada. However, populations of *P. m. rufinus*, living in desert habitat have climbing abilities that are not significantly different from those in conifer woodlands. This study shows that natural selection may not lead to the evolution of adaptive climbing behaviour in all populations, and therefore, species. It also shows that, in these subspecies, there is a lack of relation between foot length and climbing abilities.

A great number of rodents mainly move on the ground in the Saharan desert, which in its evolution has lost a good deal of shrubby vegetation. Some species are found in rocky environments but typical arboreal forms are scarce.

In the Saharan desert, *Acomys cahirinus* (Desmarest 1819) is often seen climbing the branches used to make the roof frames of cabins and *Gerbillus campestris* (Le Vaillant 1867) frequently climb branches in order to have access to cereal reserves in granaries. It seems that in Mزاب country (Algeria), some gerbils such as *Gerbillus nanus* (Blanford 1875) climb tomato-plants and eat their fruits (LE BERRE personal observation).

The aim of this study was to compare the potential locomotor adaptability of four Saharan species, in laboratory conditions, using experimental situations. Two questions were investigated: Are these species able to use the vertical dimension when it is available? Does each species behave efficiently in a complex environment?

METHODS

Biological material

The species studied were three gerbillids and one murid that occur in the Saharan desert.

Meriones shawi (Rozet 1833) a species only found in the northern part of the Sahara where it lives in plains, especially in cereal farming areas (head-body length = HBL = 140-200 mm, hind foot length = HFL = 36 mm).

Gerbillus pyramidum (L. Geoffroy 1825) a species mainly located in sandy areas (wadis beds, maaders, nebkas). It is found throughout the Saharan desert from the Red Sea (Egypt) and Saharan Atlas (Algeria) to Niger (HBL = 120, HFL = 36).

G. campestris which is found in rocky areas or earthy areas but also in human habitations (HBL = 100, HFL = 26).

The murid is *A. cahirinus* which is found in rocky areas, mainly in Egypt, Eastern and Northern Sudan, in the Djebel Uweinat (Lybia), Tassili n'Ajjer and Hoggar (Algeria). It is also found in human dwellings and granaries where it moves with ease in the roof frames (HBL = 108, HFL = 20) (LE BERRE 1990).

There are no striking morphological differences between these four species except that the hindfeet of gerbils are, in proportion to body size, longer than those of the murid. None of these species is typically arboreal.

The animals were kept in cages (40 × 20 cm) and provided, *ad libitum*, with tap water and UAR A03 mouse chow. They lived on a 14-hr light/10-hr dark cycle. The ambient temperature was $22 \pm 2^\circ\text{C}$. Twenty adult subjects, 10 males and 10 females, of each species, were tested.

All the subjects were born in the laboratory and were, at least, the fourth generation issued from wild subjects caught in the Saharan desert in 1985. They have been bred in laboratory conditions since then and all the subjects have had the same ontogenic experience in standard laboratory cages.

Laboratory tests

The locomotor abilities were studied by means of three experimental tests always performed in the same order: behaviour on a platform, on a thread and on a tree-branch. Each subject was tested once in each experiment.

The platform was a square wooden board (12×12 cm) supported by a round column (diameter: 25 mm), 15 cm high (or 20 cm for *M. shawi* which is a larger species). Each subject, held by its tail, was placed in the middle of the platform. The experimenters observed how the animal escaped from this situation: either in a passive way by falling, which is somewhat related to loss of equilibrium, or in an active way by sliding down the column or by jumping from the platform.

The thread was a 40 cm long horizontal thread 4 mm in diameter, tightened between two vertical supports. The subject was handled by its tail, its forelegs put in contact with the middle of the thread. Its behaviour was observed: falling, grasping without moving along the thread or grasping and moving along it, using its legs and sometimes its tail.

The branch, divided in its upper end in two horizontal twigs (diameter 4 cm), was 80 cm long (diameter 12 cm at the lower end) with a slope of 45°.

The subject, held by the tail, was placed on one of the horizontal upper ends. Five types of behaviour could occur: falling, jumping without any previous exploration, immobilisation (in that case, the observation was stopped after 2 min), walking on the two horizontal upper ends, moving downwards along the branch.

Statistical methods

In a first approach, the raw data corresponding to success or failure in each category of performances for each test was coded as 0 (failure) or 1 (success). This method led to a table with 10 columns (behavioural performances) and 80 rows (subjects). A factorial analysis of correspondences (FAC), which takes into account all the variables and subjects, was run on this table (HILL 1974, LEBART 1974, LEBART et al. 1979).

Chi square tests using BROWN's algorithm (1974) were run on each experimental test in order to show differences between species and genders.

RESULTS

The FAC permits us to have an overall evaluation of the behaviour of the four species.

The first factor (32% of the total variance) opposes falling from the branch (absolute contribution = AC = 0.139) and from the thread (AC = 0.129) (negative coordinates) to descent from the platform (AC = 0.210), descent from the branch (AC = 0.208) and moving along the thread (AC = 0.129) (positive coordinates). This axis could be interpreted as discriminating the performances of species which are typically terrestrial, *G. pyramidum* and *M. shawi* (negative coordinates), from those of less terrestrial species, *A. cabirinus* (positive coordinates).

The second factor (17% of the total variance) opposes jumping from the platform (AC = 0.257) and the branch (AC = 0.370) with negative coordinates to falling from the platform (AC = 0.238) with positive coordinates. It could be interpreted as discriminating the active (jumping) from the passive reactions (falling) in a three dimensional situation. It is mainly due to a few individuals of *G. campestris* (negative coordinates), *G. pyramidum* and *M. shawi* (positive coordinates).

Table 1, which summarizes the behavioural performances of the four species in the three experimental tests, shows that on the platform, *Acomys* is characterized by descent along the column which is significantly different from random ($\chi^2 = 68.28$,

Table 1.
Results of the three types of tests for each species.

	Platform			Thread			Branch			
	1	2	3	4	5	6	7	8	9	10
<i>Meriones shawi</i>	14	6	0	18	2	0	10	0	9	1
<i>Gerbillus pyramidum</i>	7	13	0	16	4	0	15	0	5	0
<i>Gerbillus campestris</i>	4	11	5	2	6	12	2	2	0	16
<i>Acomys cahirinus</i>	0	1	19	4	9	7	0	0	0	20

Frequencies of behavioural occurrences in each species: platform (1 = fall, 2 = jump, 3 = descent); thread (4 = fall, 5 = grasping, 6 = moving along); branch (7 = fall, 8 = jump, 9 = equilibrium, 10 = walking down).

df = 6, $P < 0.05$). Most *M. shawi* fell from platform ($\chi^2 = 19.81$, df = 5, $P < 0.05$). Most *G. pyramidum* and *G. campestris* escaped by jumping and there is no significant difference between these two species.

On the thread, *Acomys* ($\chi^2 = 30.00$, df = 5, $P < 0.05$) and *G. campestris* ($\chi^2 = 46.7$, df = 6, $P < 0.05$) fell significantly less than expected. Some of them succeeded in grasping the thread in their forelegs and some of them moved along it; *G. campestris* used its tail to increase its contact with the thread. *G. pyramidum* (which showed significantly more grasping, $\chi^2 = 9.54$, df = 4, $P < 0.05$, than expected) and *M. shawi* failed in this task.

On the branch the same two pairs of species were similarly differentiated.

From these data it appears that *A. cahirinus* and *G. campestris* responded actively to these situations whereas *M. shawi* and to a lesser extent *G. pyramidum* were unable to cope with the situations. No significant differences were found between the performances of the two sexes.

DISCUSSION

These tests show different behavioural performances for the four species. Two species (*A. cahirinus* and *G. campestris*) were able to move in a complex environment without falling. The two other species (*M. shawi* and *G. pyramidum*) had a reduced and inadequate control of mobility in this environment.

The differences in climbing abilities noted here are in agreement with those of THOMPSON (1990), in that large foot length is not related to greater climbing performances. Our results can be partly related to the ecological characteristics of the

environment of the four species. *A. cabirinus* and *G. campestris* live in the wild in environments where they can climb branches. Small trees and shrubs, even if they are scarce, are not completely absent in the habitat of *M. shawi* and *G. pyramidum* but these animals do not seem to climb in them, a fact perhaps related to different food seeking strategies.

From a global point of view, gerbils are adapted to open environments (open deserts and steppes). Their locomotion in these environments consists of quadruped running sometimes interrupted by biped jumps which increase the speed of locomotion.

The presence of the murid *Acomys* in the mountainous environments of the central part of Sahara could be interpreted as a remnant of an earlier settlement of rodents in a period when humidity permitted the development of a marked arboreal vegetation.

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